VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY
Chennai, India · Office of International Relations · Value Added Course

Digital Twin with Python

VTR UGE 21

DAY 5 · MODULE 5

Visualization and Capstone Development

A dashboard for the operator, maturity computed from the state of the twin, verification and validation, a credibility scorecard and the assembled twin under a fault.

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Overview

The last day assembles the twin and asks whether it deserves trust. A dashboard shows the operator what matters, including how far the twin itself can be trusted. The maturity level of Day 1 is computed from the state of the twin instead of being declared. A frozen configuration and an event log make every run reproducible [1]. Verification checks that the equations are solved correctly, validation checks that they are the right equations [2], and a credibility scorecard collects the evidence [3]. The assembled twin is finally run on a healthy and on a faulted pack.

Estimated study time: 6 to 8 hours.

Learning outcomes

By the end of the day, students are expected to build an operator dashboard that shows state, trust and alarms, to freeze a configuration and attach its fingerprint to every logged record, to compute the maturity level of a twin from its data flows and its trust in itself, to verify code and solution numerically, to validate a model and compute the model error implied above the noise of the sensor, to fill a credibility scorecard with evidence and open items, and to run an assembled twin through a fault and interpret its fail-safe behaviour. In Python, students are expected to write classes and frozen dataclasses, to log records as JSON lines, one record per line in the JavaScript Object Notation, and to read them back with pandas.

Python in this day

The Colab notebook opens with Python step 5: Classes, records and files, right after its setup section. It covers frozen dataclasses with a fingerprint method, a class that writes JSON lines to a file, and reading the log back into a data frame, applied to the configuration and the event log of the twin [4, 5]. After the step, the notebook continues with the assembled twin.

How these notes point to the Python code. Every method of this lecture is also written as Python code in the Colab notebook of the day, the file `NB05_visualization_and_capstone.ipynb`. A Colab notebook is a document of text and Python code that runs in the web browser, with nothing to install. It opens with the button Open in Colab on the day overview page of the course, https://github.com/utkukose/digital-twin-veltech-NB-lecture/tree/main/days/day-05. The code of the notebook is divided into numbered sections. In these notes, each box titled *Python code in the Colab notebook, Section N* names the section whose code carries out what the text above the box explains, and says what to run and what to look at there. The animations that the notes mention are interactive figures on the lecture page of the day, https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lecture.html. The interactive lab, with the practice parts and the self-assessment of the day, is at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html.

Python step 5: Classes, records and files

A twin that runs for months must be able to show which settings produced which result. This step freezes a configuration in a dataclass with a fingerprint, writes an event log with a small class and reads it back with pandas [4, 5]. A class describes one kind of object: the data that every such object holds, called attributes, and the functions that belong to it, called methods.

A frozen configuration with a fingerprint

```
from dataclasses import dataclass, asdict
import json, hashlib

@dataclass(frozen=True)
class TwinConfig:
    dt_s: float = 10.0
    sync_period_s: float = 60.0
    fan_on_C: float = 42.0
    seed: int = 42

    def fingerprint(self):
        text = json.dumps(asdict(self), sort_keys=True)
        return hashlib.sha256(text.encode()).hexdigest()[:12]

cfg = TwinConfig()
print(cfg)
print("fingerprint:", cfg.fingerprint())
```

Output

```
TwinConfig(dt_s=10.0, sync_period_s=60.0, fan_on_C=42.0, seed=42)
fingerprint: 198bfb885b09
```

`@dataclass(frozen=True)` creates a class whose attributes cannot be changed after creation. A dataclass lists its attributes with their types and default values, and `cfg = TwinConfig()` creates one object of the class. The method `fingerprint` turns the content into sorted JSON text and hashes it, so the same settings always give the same fingerprint. JSON, the JavaScript Object Notation, is a plain-text format of named values. A hash function turns a text of any length into a fixed number of characters that changes completely when the text changes. Here it is SHA-256, the Secure Hash Algorithm with an output of 256 bits, of which the first 12 characters are kept.

A frozen object refuses changes

```
try:
    cfg.fan_on_C = 40.0
except Exception as err:
    print("refused:", type(err).__name__)
cfg2 = TwinConfig(fan_on_C=40.0)
print("one setting changed, new fingerprint:", cfg2.fingerprint())
```

Output

```
refused: FrozenInstanceError
one setting changed, new fingerprint: 6208baa01baf
```

`try` runs a block and `except` catches the error it raises, here the refusal of the frozen object. A changed setting requires a new object, and the new object has a different fingerprint, so the change cannot go unnoticed.

A class that writes a log

```
class TwinLog:
    def __init__(self, path, cfg):
        self.path, self.cfg = path, cfg
        open(self.path, "w").close()          # start an empty file

    def write(self, **record):
        record["config"] = self.cfg.fingerprint()
        with open(self.path, "a") as fh:
            fh.write(json.dumps(record) + "\n")

log = TwinLog("twin_log.jsonl", cfg)
for t, T in [(0, 36.0), (10, 36.2), (20, 36.5)]:
    log.write(t=t, T_belief=T, fan=0.0)
print(open("twin_log.jsonl").read())
```

Output

```
{"t": 0, "T_belief": 36.0, "fan": 0.0, "config": "198bfb885b09"}
{"t": 10, "T_belief": 36.2, "fan": 0.0, "config": "198bfb885b09"}
{"t": 20, "T_belief": 36.5, "fan": 0.0, "config": "198bfb885b09"}
```

`**record` collects any keyword arguments into a dictionary, so `write` accepts records with any fields. `with open(self.path, "a") as fh:` opens the file under the name `fh` and closes it again when the indented block ends. The mode `"a"` appends to the file, and every record becomes one line of JSON text.

Reading the log back

```
import pandas as pd
back = pd.read_json("twin_log.jsonl", lines=True)
print(back)
print("distinct configurations:", back["config"].nunique())
```

Output

```
    t  T_belief  fan        config
0   0      36.0    0  198bfb885b09
1  10      36.2    0  198bfb885b09
2  20      36.5    0  198bfb885b09
distinct configurations: 1
```

`pd.read_json(..., lines=True)` reads one record per line into a data frame, the table of pandas with named columns, and `nunique` counts the distinct values of a column. One distinct fingerprint means that the whole log was produced by one configuration.

Quick check. Why is the configuration of the twin frozen?

A dashboard for the operator

> **The twin of the pack so far**
>
> Over four days the twin of the pack has gained its parts. Day 1 stated the model and the two data flows of a twin, from the sensor to the belief and from the belief to the fan. Day 2 verified the simulation core. Day 3 cleaned the telemetry, calibrated the parameters and corrected the belief with a filter. Day 4 added the services: surrogates, a hybrid model, fault detection with a cumulative sum, the remaining useful life and a controller that plans the fan.
>
> This day puts the parts of the project together and asks the question that decides whether the twin may be used: how far it deserves trust. It ends with the brief of the final project, in which every student builds a twin of this kind for an asset of their own.

An operator needs few numbers, at the right time, with their meaning. Three things belong on one screen: the state of the asset, how far that state can be trusted, and the alarms that require action. The dashboard of the project shows the temperature against its limit, the fan duty and one line that is often missing from twin dashboards: the residual between the twin and the sensor, which tells the operator how far the twin can be trusted at this moment. A good dashboard uses colour sparingly, for states that require action, and shows units on every axis.

> **In everyday life: a dashboard**
>
> The dashboard of a car shows few numbers: the speed, the fuel and a handful of warning lights that come on only when the driver has to act. The car measures far more, and showing all of it would make the car harder to drive. The dashboard of a twin follows the same rule and adds one line that a car rarely has: how far its own picture of the asset can be trusted.

The asset and its model. The asset of the course is the 48 V lithium-ion battery pack of Day 1. The current I of its load heats it through the internal resistance R with the power $I^2 R$, the surrounding air cools it, and a fan adds to the cooling. The temperature T of the pack follows the balance of the two:

$$\frac{dT}{dt} = \frac{I^2 R - (hA_0 + hA_1 u)\,(T - T_{\text{amb}})}{mc_p}$$

The left side is the rate of change of the temperature, in degrees per second. T_{amb} is the ambient temperature, the temperature of the surrounding air, 35 °C on the trips of this day. The fan duty u runs from 0, fan off, to 1, full speed. The conductances hA_0 and hA_1, in W/K, state how many watts leave the pack for each degree above the ambient temperature: hA_0 without the fan, and hA_1 more at full fan. The heat capacity $m\,c_p$, in J/K, is the energy that warms the pack by one degree. The twin never sees the values of the pack. It works with the values that Day 3 calibrated from measurements, which differ from the true ones by up to 2 percent.

Parameter	Symbol and name in the code	Simulated pack	Twin, calibrated on Day 3
Internal resistance	R, `R0`	0.045 ohm	0.0456 ohm
Heat capacity	$m\,c_p$, `m_cp`	9000 J/K	9180 J/K
Conductance without the fan	hA_0, `hA0`	2.2 W/K	2.193 W/K
Conductance added by the full fan	hA_1, `hA1`	12.0 W/K	12.23 W/K

> **Python code in the Colab notebook, Section 1.**
>
> Open Section 1 of the Colab notebook and run it. The cell defines the two parameter sets of the table above, `TRUE` for the simulated pack and `CAL` for the twin, and prints the calibrated values. The model itself, the function `pack_derivatives`, stands in Section 0, the setup.

State. Besides the temperature, the dashboard shows the state of charge, abbreviated SoC, which is the share of the capacity Q still in the pack. The full pack holds 60 Ah when new, and the current empties it at the rate

$$\frac{d\,\mathrm{SoC}}{dt} = -\frac{I}{3600\,Q}$$

The factor 3600 converts the hours of the capacity into seconds. A current of 40 A empties a full pack in 60 / 40 = 1.5 hours. The capacity itself fades with use, but over months, so it hardly changes within one trip.

The loop of the twin. The pack and its twin are simulated side by side for 90 minutes of a highway trip, in which the current swings between 34 and 46 A. Both start at 36 °C and advance in steps of Δt = 10 s with the fourth-order Runge-Kutta method of Day 2, abbreviated RK4. The twin computes with the calibrated parameters, so the temperature that it holds, written $\hat{T}$ with a hat and called its belief, drifts slowly away from the true one. Once per minute the sensor delivers a reading y_k, where the index k numbers the readings:

$$y_k = T(t_k) + b + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, \sigma^2)$$

→ *This formula with numbers, step by step:* Lab Part A, *steps 1 to 12, one reading in each.*

The reading is the true temperature at the time t_k plus two errors. The bias b is a constant offset of the sensor, zero unless a fault is staged. The noise ε_k is drawn anew for every reading, normally distributed with mean 0 and standard deviation σ. The sensor of the course has σ = 0.35 °C, rounds its readings to steps of 0.25 °C and loses 2 percent of them. The twin compares each reading with its belief and corrects the belief:

$$r_k = y_k - \hat{T}$$

$$\hat{T} \leftarrow \hat{T} + K\,r_k$$

→ *These formulas with numbers, step by step:* Lab Part A, *steps 1 to 12, one reading in each.*

The residual r_k is the reading minus the belief at that moment. The belief moves towards the reading by the share K of the residual. K is called the gain and is 0.5 here. A belief of 44.0 °C and a reading of 44.6 °C give a residual of 44.6 − 44.0 = 0.6 °C and a new belief of 44.0 + 0.5 × 0.6 = 44.0 + 0.3 = 44.3 °C. With K = 1 the belief would copy every reading with its noise, as the shadow of Day 1 did. A rule of this form is called an observer, and the Kalman filter of Day 3 is an observer that computes its gain instead of fixing it [6]. The second data flow leads back to the pack: The twin sets the fan from its belief,

$$u = \mathrm{clip}\left(\frac{\hat{T} - T_{\mathrm{on}}}{6},\ 0,\ 1\right)$$

→ *This formula with numbers, step by step:* Lab Part A, *steps 1 to 12, one reading in each.*

The fan starts at the threshold T_{on} = 42 °C and reaches full speed 6 °C higher, at 48 °C, so a belief of 45 °C gives a duty of (45 − 42) / 6 = 3 / 6 = 0.5. The pack counts as overheated above its limit of 46 °C.

Trust. While the model describes the pack, the residuals are small and scatter around zero. A single large residual may be noise, so the twin adds the residuals up in the cumulative sum of Page that Day 4 introduced [7]:

$$S_k = \max(0,\ S_{k-1} + |r_k| - \kappa), \qquad S_0 = 0$$

→ *This formula with numbers, step by step:* Lab Part A, *steps 1 to 12, one reading in each.*

S_k is the sum after reading k. The absolute value lets residuals of both signs count. The allowance κ is subtracted at every reading and absorbs ordinary noise, and the max keeps the sum from falling below zero. The residual alarm is raised when the sum exceeds the decision limit h. The twin uses $\kappa = 0.6$ °C and $h = 5.0$ °C. Day 4 added the residuals with their sign, which detects a drift in one direction. The absolute value used here reacts to a drift in either direction.

> **Worked example: a cumulative sum by hand**
>
> Six readings have the residuals 0.4, −0.9, 0.2, 2.0, 2.4 and 2.6 °C, and the allowance is 0.6 °C. After the first reading the sum is max(0, 0 + 0.4 − 0.6) = 0. The second residual counts with its size, |−0.9| = 0.9, and gives 0 + 0.9 − 0.6 = 0.3, and the third max(0, 0.3 + 0.2 − 0.6) = 0: Small residuals are forgotten.
>
> The last three residuals are large. The sum becomes 0 + 2.0 − 0.6 = 1.4, then 1.4 + 2.4 − 0.6 = 3.2, then 3.2 + 2.6 − 0.6 = 5.2. It exceeds the decision limit of 5.0 at the sixth reading, where the alarm is raised. A limit of 3 °C on single residuals would have stayed silent.

> **Interactive lab, Part A.**
>
> The trust loop of the twin over twelve readings of a pack with a hidden fault: the residual, the cumulative sum, the corrected belief and the fan command with its fail-safe switch. Sliders set the fault, the gain, the allowance and the decision limit. Part A opens at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html#pa.

A simulation knows the true temperature, so the error of the belief can also be measured directly. Its measure is the root mean square error, abbreviated RMSE, of N errors $e_1, \ldots, e_N$:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} e_k^2}$$

The errors are squared, so that their signs do not cancel, then averaged, and the root returns to degrees: Errors of 0.1, −0.2 and 0.2 °C have the squares 0.01, 0.04 and 0.04, which add up to 0.09, and an RMSE of √(0.09 / 3) = √(0.03) = 0.17 °C. On the dashboard, e_k is the belief minus the true temperature at each of the $N = 540$ steps. The trust reads `OK` for a belief RMSE below 1 °C, `DEGRADED` up to 3 °C and `DO NOT TRUST` above. On a real asset the true temperature is unknown and this number cannot be computed, which is why the alarm is built on the residuals.

Alarms. The dashboard has two alarms. The overheat alarm, a temperature above the limit of 46 °C, concerns the asset. The residual alarm of the cumulative sum concerns the twin.

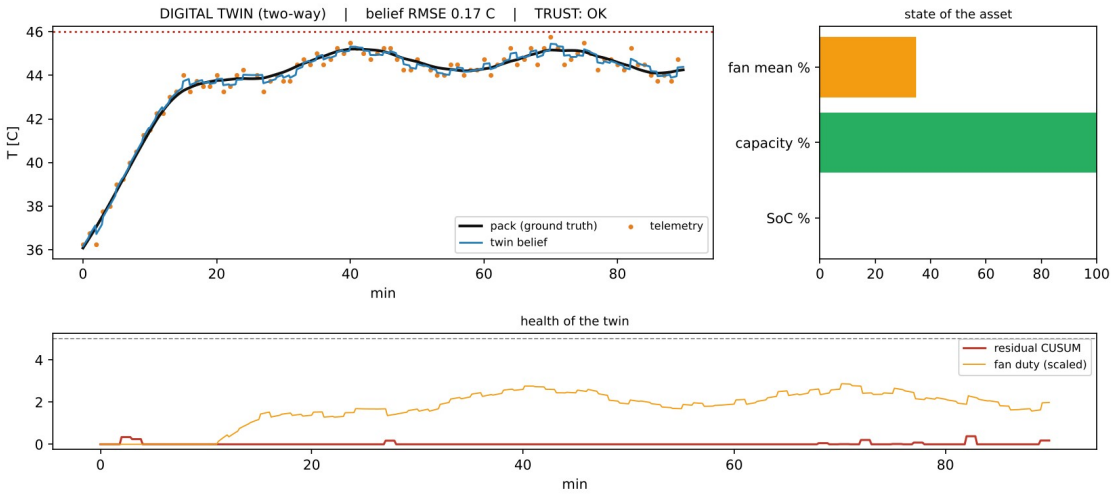


Figure. The dashboard of the twin after a healthy run of 90 minutes: temperature with its limit, the state of the asset, and the cumulative sum of the residuals that carries the trust of the twin.

In the upper left panel of the figure the black line is the true temperature of the pack, the blue line the belief of the twin and the orange dots the readings. The dotted line marks the limit of 46 °C. The fan starts at minute 11, when the belief passes 42 °C, and the temperature levels off near 44 to 45 °C, with a peak of 45.21 °C. The title gives the level of the twin, the belief RMSE of 0.17 °C and the trust `OK`. The bars on the right summarise the state of the asset. The fan ran at 35 percent on average, and the capacity is at 99.96 percent of its new value. The state of charge is at zero. The pack started at 92 percent, that is with 0.92 × 60 = 55.2 Ah, and a mean current of 40 A draws this charge in 55.2 / 40 = 1.38 hours, which is 1.38 × 60 = 82.8, about 83 minutes. In the simulation, in which the current swings, the pack is empty after 82 minutes. The lower panel shows the health of the twin. The red line, the cumulative sum, never rises above 0.4 and stays far below its decision limit of 5, the dashed line. The orange line is the fan duty, multiplied by 5 to share the axis. The pack spent 0 s above 46 °C, and no residual alarm was raised.

Python code in the Colab notebook, Section 2.

Open Section 2 of the Colab notebook and run it. The function `run_twin` simulates the pack and its twin with the observer and the cumulative sum, and `dashboard` draws the figure shown here and prints the peak temperature, the seconds above 46 °C and the time of the residual alarm. In Colab the cell shows sliders above the figure. Move `T_amb` and `fan_on` to change the day and the fan threshold, untick the box `twin may actuate` to turn the twin into a shadow, and move `sensor_bias` to stage a faulty sensor.

Check your understanding (1). The cumulative sum of a twin stands at 4.2 °C, with an allowance of 0.6 °C and a decision limit of 5.0 °C. The next residual is −1.5 °C. What happens?

A. The sum falls to 2.1 °C and no alarm is raised

B. The sum rises to 5.1 °C and the alarm is raised

C. The sum rises to 5.7 °C and the alarm is raised

D. The sum is reset to zero

Maturity computed from the state of the twin

Day 1 defined models, shadows and twins by their data flows [8]. A digital model exchanges no data with the asset automatically, a digital shadow receives data from the asset automatically, and a digital twin also sends commands back automatically. A running system can lose a flow at any moment: The telemetry may stop, the actuator may be disabled, or the belief may drift beyond what the residual allows. The twin of the project therefore computes its maturity level from its live state, so that the label shown to the operator is always true.

Telemetry is the stream of readings that the asset sends, and the actuator is the device through which the twin acts on the asset, here the fan. The computation takes three facts: whether telemetry arrives, whether the actuator responds, and the RMSE of the belief. It checks the rules of the table from top to bottom, and the first rule that applies decides.

Rule	Condition	Level	Meaning
1	No telemetry arrives	`DIGITAL MODEL`	Open loop: The predictions are not anchored to the asset and must not be acted on
2	The belief RMSE exceeds the trust limit of 3.0 °C	`DIGITAL MODEL (degraded)`	Telemetry is present, but the model no longer follows it
3	The actuator does not respond	`DIGITAL SHADOW`	Monitoring only: The twin cannot influence the asset
4	None of the above	`DIGITAL TWIN`	Closed loop: The twin acts on the asset

The order of the rules carries the meaning. Without telemetry the twin is a model, whatever its actuator can do. With a belief that is off by more than the trust limit, the status is degraded even when both flows work. A twin whose model no longer describes the asset is no longer a twin, and it should say so and fall back to a safe action. Five test cases show the rules at work. With both flows the level is `DIGITAL TWIN` for a belief RMSE of 0.4 °C and `DIGITAL MODEL (degraded)` for 5.2 °C. Without actuation it is `DIGITAL SHADOW`, and without telemetry `DIGITAL MODEL`.

> **In everyday life: a status that is computed**
>
> A phone shows no bars or the words emergency calls only when it loses its network, instead of the full signal it had an hour ago. Its status is computed from the live link, not copied from the brochure. A twin that labels itself in the same way never shows an operator a promise that the running system no longer keeps.

In the animation both flows can be switched off and on. Its trust limit is a slider that starts at 1.5 °C, and it shows the degraded state as `DEGRADED`. Its last line names the policy of each state: Only the digital twin controls the fan, and the degraded state switches to fail-safe cooling, the fan at full speed.

Animation on the lecture page: The maturity level computed from the state of the twin. Switch telemetry and actuation off and on and raise the belief error above the trust limit.

Python code in the Colab notebook, Section 3.

Open Section 3 of the Colab notebook and run it. The function `maturity_level` is the table above, one `if` for each rule. The cell calls it for five combinations of telemetry, actuation and belief RMSE and prints the level and its meaning for each.

Check your understanding (2). The telemetry of a twin stops for ten minutes. What should the dashboard show?

A. Digital twin, as designed

B. A lower level, because the automatic flow from the asset is missing

C. Nothing

D. An error

Configuration and the event log

A result that cannot be reproduced cannot be checked. The twin of the project therefore stores every setting of a run in a frozen configuration with a fingerprint and writes every step, with its belief and its command, to an event log [1]. Two runs with the same configuration produce the same log, and a changed setting changes the fingerprint.

In everyday life: a record that can be checked

The flight recorder of an aircraft stores the settings and the measurements of every second with their times, so that investigators can reconstruct a flight long after it ended. A recipe with exact amounts, times and temperatures does the same for a cake: Another cook can repeat it and get the same result. The frozen configuration and the event log make a run of the twin repeatable and checkable in this sense.

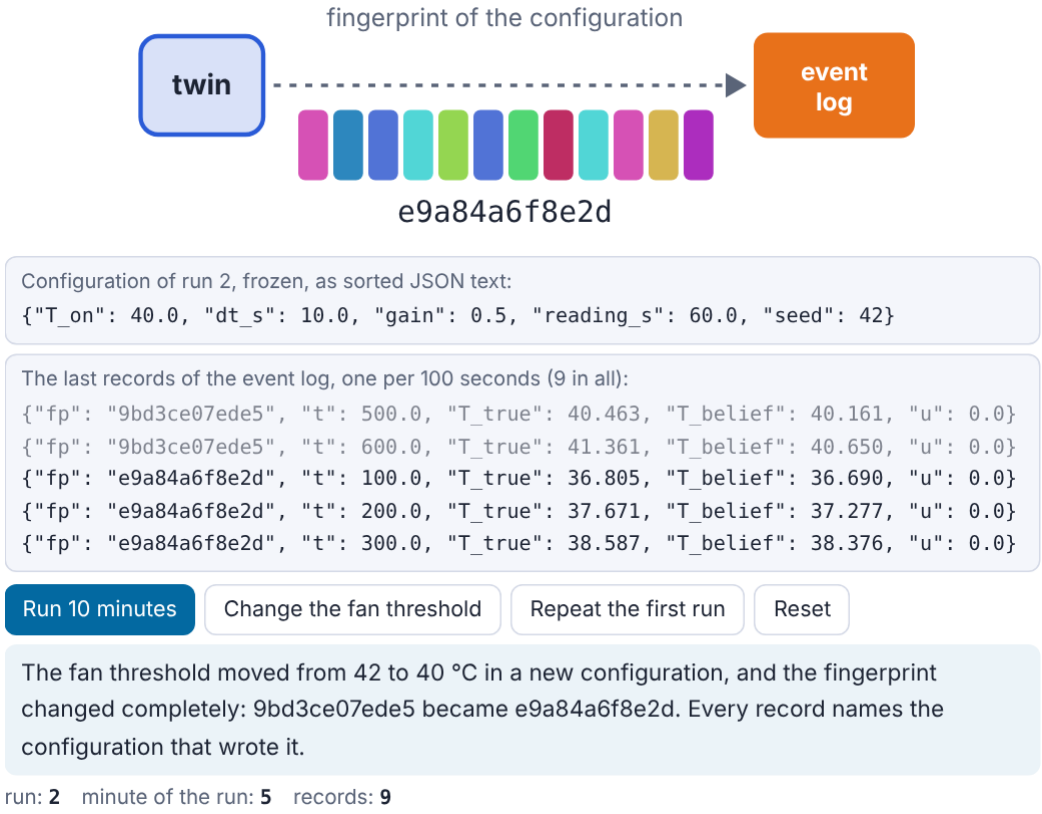
The frozen configuration. A configuration is the complete list of settings that decide how a run behaves. The configuration of the twin holds 15 of them. Among them are the time step, the interval between readings, the fan threshold, the gain of the observer, and the allowance and the decision limit of the cumulative sum. The four parameters of the model and the seed belong to it as well. The seed is the number that starts the random number generator: With the same seed the simulated sensor produces the same noise. Frozen means that no setting can be changed after the object has been created, so a different setting requires a new configuration.

The fingerprint. The fingerprint is a short code computed from the settings. They are first written as text in JSON, the JavaScript Object Notation, a plain-text format of named values, with the names in sorted order. A hash function then turns the text into a fixed number of characters. The twin uses SHA-256, the Secure Hash Algorithm with an output of 256 bits, written as 64 hexadecimal characters, and keeps the first 12:

$$\text{fingerprint} = \text{first}_{12}\,(\text{SHA256}\,(\text{settings as sorted JSON text}))$$

The same settings always give the same fingerprint, `dea2275ffa75` for the configuration of the day. A single changed setting gives an unrelated one: With the fan threshold at 40 °C instead of 42 °C the fingerprint is `a54d02475539`. The settings cannot be read back from a fingerprint, so the configuration itself is saved as a JSON file and can be loaded again unchanged.

The event log. An event log is a file to which the twin only appends, one record per step. The twin writes JSON lines, one record per line, for example `{"fp": "dea2275ffa75", "t": 100.0, "T_true": 36.888, "T_belief": 37.023, "u": 0.0}`. The field `fp` is the fingerprint, `t` the time in seconds and `u` the fan duty. The fingerprint stands in every record, because logs are split, merged and partly lost, and a record that cannot name its configuration proves nothing. The true temperature can be logged only because the pack is simulated.



Interactive scene on the lecture page: The event log. The twin of the pack writes its event log, one record per 100 s, each with the fingerprint of its configuration. The scene uses five of the settings and computes the fingerprint as the notebook does: the first 12 characters of the SHA-256 hash of the settings as sorted JSON text. Changing the fan threshold needs a new configuration and gives an unrelated fingerprint, and repeating the first run writes the same records, digit for digit.

The log of one trip keeps every tenth step, one record per 100 s: 54 records in 4605 bytes, about 85 bytes per record. The storage grows with the number of records:

$$\text{bytes per day} = \frac{86400}{\text{seconds per record}} \times \text{bytes per record}$$

A day has 86400 seconds, so one record per 100 s gives 86400 / 100 = 864 records, and with 4605 / 54 = 85.3 bytes per record about 864 × 85.3 = 73 700 bytes per day for one asset. Counted with 1024 bytes per kilobyte and 1024 kilobytes per megabyte, this is 73 700 / 1 048 576 = 0.07 megabytes, about 0.1 megabytes per day, to be multiplied by the size of the fleet.

Worked example: the log of a fleet

A fleet of 200 assets logs one record of 100 bytes every 10 s. One asset writes 86400 / 10 = 8640 records per day, that is 864 000 bytes or 0.864 megabytes of one million bytes. The fleet writes 200 × 0.864 = 172.8 megabytes per day, and 172.8 × 365 = 63 072 megabytes, about 63 gigabytes, per year.

Sent over a network, the same stream needs 200 × 100 × 8 / 10 = 16 000 bits per second, since a byte has 8 bits. This data rate is what the connection must carry, and it is called the required bandwidth.

Python code in the Colab notebook, Section 4.

Open Section 4 of the Colab notebook and run its two cells. In the first cell the class `TwinConfig` is the frozen configuration with its method `fingerprint`. The cell saves the configuration to a JSON file, loads it again and prints the fingerprint, the result of the comparison and the fingerprint after one changed setting. In the second cell the class `TwinLog` writes the event log of one run, and the cell prints the number of records, the size of the file and the storage per day and per year.

Verification and validation

Verification asks whether the equations are solved correctly: Halving the step must reduce the error by the factor that the order of the method predicts [2]. Validation asks whether they are the right equations: The twin must match measurements it has not seen. Validation has a floor. The error between twin and sensor can never fall below the noise of the sensor, so the error of the model itself is estimated by subtracting the noise, and small model errors can hide below the noise.

In everyday life: right answer, right question

A spell checker verifies a letter: It finds every word that is not spelled correctly. It cannot validate the letter, because a letter without a single typo can still say the wrong thing to the wrong person. Verification of a twin asks whether its program solves its equations correctly, validation whether the equations describe the asset.

Interactive lab, Part B.

Six checks of a twin, each to be sorted into verification or validation. Part B opens at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html#pb.

Verification has two parts, code verification and solution verification. Neither needs a measurement of the asset: Both compare numbers of the program with other numbers. Unit tests, small automatic checks of single functions, belong to verification as well.

Code verification. Code verification runs the program on a problem whose exact solution is known. For a constant current and a constant fan duty the energy balance has one, the exponential approach to a final value that Day 2 used:

$$T(t) = T_\infty + (T_0 - T_\infty)\, e^{-t/\tau}$$

$$T_\infty = T_{\text{amb}} + \frac{I^2 R}{hA_0 + hA_1 u}$$

$$\tau = \frac{mc_p}{hA_0 + hA_1 u}$$

T_0 is the temperature at the start. T_∞ is the steady temperature at which heating and cooling balance. The time constant τ sets the pace: After one time constant, 63 percent of the way to T_∞ is covered.

> **Worked example: the exact solution of the test case**
>
> The test case of the project has T_0 = 36 °C, a current of 40 A, a fan duty of 0.3 and the calibrated parameters. The heating power is $40^2 \times 0.0456 = 1600 \times 0.0456 = 72.96$ W, and the conductance is 2.193 + 12.23 × 0.3 = 2.193 + 3.669 = 5.862 W/K. The steady temperature is 35 + 72.96 / 5.862 = 35 + 12.45 = 47.45 °C, and the time constant is 9180 / 5.862 = 1566 s, which is 1566 / 60 = 26.1, about 26 minutes. In this and the following worked examples, intermediate values are rounded, and each step continues with the rounded value.
>
> After one hour, t = 3600 s, the exponent is 3600 / 1566 = 2.299, and the exponential factor is $e^{-2.299}$ = 0.1004. The distance at the start is 36 − 47.45 = −11.45 °C, so the exact temperature is 47.45 − 11.45 × 0.1004 = 47.45 − 1.15 = 46.30 °C. Every numerical result for this hour is compared with this number.

The integrator of the course solves the same hour in steps of Δt. It is the RK4 method of Day 2, which combines four slopes of the model in each step. The difference between its result and the exact one is the error $e(\Delta t)$. For a method of order p the error is proportional to the p-th power of the step, and the order that a program really shows, the observed order, follows from two errors:

$$e(\Delta t) \approx c\, \Delta t^{\,p}$$

$$p_{\text{obs}} = \log_2 \frac{e(\Delta t)}{e(\Delta t/2)}$$

The constant c depends on the problem. Halving the step divides the error by 2^p: by 2 for the explicit Euler method, whose order is 1, and by 16 for RK4, whose order is 4. The table shows the errors for this hour.

Step Δt	Steps in one hour	Error $e(\Delta t)$ in °C	Observed order from this row and the next
400 s	9	1.16×10^{-4}	4.15
200 s	18	6.51×10^{-6}	4.08
100 s	36	3.86×10^{-7}	4.04
50 s	72	2.35×10^{-8}	4.02
25 s	144	1.45×10^{-9}	

The first two rows have the ratio $1.16 \times 10^{-4} / (6.51 \times 10^{-6}) = 17.8$, which gives an observed order of $\log_2 17.8 = \ln 17.8 / \ln 2 = 2.879 / 0.693 = 4.15$. Each further halving brings the ratio closer to 16 and the order closer to 4. The verdict is `PASS` when the last value exceeds 3.5. An observed order of 1 or 2 would reveal a programming error in the integrator, even when the temperature curve looks plausible.

Solution verification. The second part asks how large the numerical error is in the simulation that is actually deployed: the highway trip of 90 minutes with the step of 10 s. The error that a finite step causes is called the discretisation error. No exact solution exists for this trip. The trip is therefore repeated with

steps from 40 s down to 2 s, with the fan off, and the final temperature is compared with a reference run at a step of 0.5 s.

Step Δt	Final temperature in °C	Difference to the reference in °C
40 s	59.175853	0.000251
20 s	59.175783	0.000321
10 s	59.175900	0.000204
5 s	59.175996	0.000108
2 s	59.176066	0.000038

At the deployed step of 10 s the discretisation error is 0.0002 °C, and 0.35 / 0.0002 = 1750: It is more than a thousand times smaller than the sensor noise of 0.35 °C. The step is fine enough, and numerical error can be ruled out as a cause of what validation finds. Two details of the table are instructive. From 10 s to 5 s the error falls by a factor of 0.000204 / 0.000108 = 1.9 and not by 16. The simulation reads the current once at the start of each step and holds it for the whole step, and this hold is only a first-order approximation. The error at 40 s is smaller than at 20 s, because at large steps two error terms of opposite sign partly cancel. A test with a constant current shows neither effect, which is why the deployed configuration is checked on its own.

Validation. Validation compares the twin with measurements of the asset, under conditions that it was and was not calibrated on. The range of conditions that the calibration data covered is called the calibration envelope. Five experiments of one hour run with the fan fixed at a duty of 0.4. The first is the highway trip at 35 °C, the condition of the calibration. The other four lie outside the envelope, with another drive cycle, a cold and a hot ambient temperature and an idle period. In each experiment the model is compared with $N = 361$ noisy readings, one every 10 s, and the errors e_k are the model temperature minus the reading.

Condition	Used in the calibration	RMSE in °C	Largest error in °C	Mean error in °C
Highway trip at 35 °C	yes	0.319	1.096	0.024
Urban trip at 35 °C	no	0.355	1.001	0.005
Highway trip at 22 °C	no	0.357	1.167	−0.024
Urban trip at 45 °C	no	0.346	1.037	−0.034
Idle at 30 °C	no	0.338	0.969	0.021

The mean errors are close to zero, so the model is not shifted against the pack. The RMSE lies between 0.32 and 0.36 °C in all five conditions, inside and outside the envelope, which is the size of the sensor noise.

The noise floor. The difference between model and reading has two sources: the error of the model, with the RMSE e_{model}, and the noise of the sensor, with the standard deviation σ. When the two are independent, their squares add:

$$\text{RMSE}^2 = e_{\text{model}}^2 + \sigma^2$$

→ *This formula with numbers, step by step:* Lab Part C, *step 1.*

A perfect model, with $e_{\text{model}} = 0$, still shows an RMSE of σ. This is the noise floor of the validation. Solving for the model error gives the implied model error:

$$e_{\text{model}} = \sqrt{\max\left(0,\ \text{RMSE}^2 - \sigma^2\right)}$$

→ *This formula with numbers, step by step:* Lab Part C, *step 2.*

The max guards against a negative number under the root, which occurs when the RMSE falls below σ by chance. The largest RMSE of the five conditions is 0.3565 °C before rounding. Its square is $0.3565^2 = 0.12709$, the square of the noise is $0.35^2 = 0.1225$, and the implied model error is $\sqrt{(0.12709 - 0.1225)} = \sqrt{(0.00459)} = 0.068$ °C.

> **Worked example: the model error behind an RMSE**
>
> A twin is validated with a sensor whose noise has $\sigma = 0.40$ °C, and the RMSE between twin and readings is 0.50 °C. The implied model error is $\sqrt{(0.50^2 - 0.40^2)} = \sqrt{(0.25 - 0.16)} = 0.30$ °C: The model is wrong by more than the sensor can hide.
>
> In the other direction, a model error of 0.10 °C under the same noise gives an RMSE of $\sqrt{(0.10^2 + 0.40^2)} = \sqrt{(0.01 + 0.16)} = \sqrt{(0.17)} = 0.412$ °C. The RMSE rises by only $0.412 - 0.40 = 0.012$ °C above the floor. Because the errors add as squares, a model error well below the noise changes the RMSE very little.

> **In everyday life: a change below the noise**
>
> A bathroom scale whose readings scatter by 200 grams cannot show that a person has gained 50 grams overnight. The change is real, but it lies below the noise floor of the scale. Weighing a hundred times and averaging would reveal it, and so does a validation with many more readings, as the next paragraph computes.

An implied model error must be read with care, because an RMSE from N readings is itself uncertain. From one experiment to the next it scatters around its long-run value with a standard deviation of about

$$\text{SD(RMSE)} \approx \frac{\text{RMSE}}{\sqrt{2N}}$$

→ *This formula with numbers, step by step:* Lab Part C, *steps 3 to 5.*

With $N = 361$ readings and an RMSE near 0.35 °C the scatter is $0.35 / \sqrt{(722)} = 0.35 / 26.87 = 0.013$ °C. The largest RMSE of the table lies $0.3565 - 0.35 = 0.007$ °C above the noise, half of this scatter, and the RMSE of the calibration condition, 0.319 °C, lies below the noise, which only chance can produce. All five values are what a perfect model would show. The validation therefore supports one statement: The model error is too small to be detected with this sensor. It does not show that the error is 0.068 °C, and it does not show that the error is zero. The simulation, which knows the true temperature, confirms this reading: The model deviates from the true pack by an RMSE of 0.014 °C at most. A model error would have to reach about 0.14 °C to lift the RMSE to $\sqrt{(0.14^2 + 0.35^2)} = \sqrt{(0.0196 + 0.1225)} = 0.377$ °C, twice the scatter above the floor, $0.35 + 2 \times 0.013 = 0.376$ °C. More readings help slowly, since four times as many only halve the scatter.

Interactive lab, Part C.

The noise floor computed step by step for a measured RMSE, a sensor noise and a number of readings of your choice: the implied model error, the scatter of the RMSE, the distance above the floor and the smallest model error that the data could show. Part C opens at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html#pc.

Interactive lab, Part D.

Three model errors behind validation results computed by hand, with the formula of the noise floor. The lab checks each value. Part D opens at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html#pd.

The animation repeats a validation 120 times for each number of samples, with a sensor noise and a true model error that the reader sets. It draws the median of the implied model error and the band that holds 95 percent of the experiments. With a noise of 0.35 °C and a true model error of 0.07 °C the lower edge of the band stays at zero up to 2000 samples. Such data cannot exclude a model error of zero. For 540 samples the readout gives a median of 0.066 °C and a band from 0 to 0.15 °C. With a true model error of 0.14 °C, or with a sensor noise of 0.15 °C, the band leaves zero.

Animation on the lecture page: The model error implied by validation data against the number of samples, from 120 repeated experiments per point. Lower the sensor noise or raise the true model error and watch when the band separates from zero.

Python code in the Colab notebook, Section 5.

Open Section 5 of the Colab notebook and run its cells. The first cell compares `rk4_step` with the exact solution `exact_T` for five steps and prints the errors, the observed orders and the verdict. It then repeats the deployed trip with five steps and prints the second table above. After a short cell that prints the last observed order, the next cell runs the five validation experiments with `validation_experiment` and prints the third table, the largest RMSE, the noise and the implied model error. Exercise A asks for an implied model error by hand.

Check your understanding (3). The error between twin and sensor equals the sensor noise. What can be concluded about the model error?

A. It is zero

B. It is too small to be detected with this sensor

C. It equals the noise

D. The model is wrong

Check your understanding (4). A twin is validated with a sensor whose noise has a standard deviation of 0.12 °C, and the RMSE between twin and readings is 0.13 °C. What is the implied model error?

A. 0.01 °C

B. 0.05 °C

C. 0.13 °C

D. 0.18 °C

The credibility scorecard

A scorecard collects the evidence for trust in one place: verification of code and solution, validation against data, the quantification of uncertainty, the documentation of the configuration and the behaviour under faults [3]. Each element is marked as passed, partial or not done and carries one line of evidence. For the twin of the course, verification and validation inside the calibrated range pass; validation outside that range reaches only the noise floor of the sensor, and the uncertainty of the inputs, the form of the model and a test on real hardware are still missing. An honest scorecard shows these gaps instead of hiding them.

Element	Status	Evidence
Code verification	`PASS`	Fourth-order convergence against an exact solution
Solution verification	`PASS`	Discretisation error of 0.0002 °C at a step of 10 s
Validation inside the envelope	`PASS`	RMSE of 0.32 °C on the calibration condition
Validation outside the envelope	`PARTIAL`	RMSE of 0.36 °C, the noise floor of the sensor: The model error cannot be resolved
Parameter uncertainty	`PARTIAL`	Intervals of the four-parameter fit are invalid under the scaling symmetry, and the heat capacity comes from a separate test (Day 3)
Input uncertainty	`NOT DONE`	Ambient temperature and current are treated as exactly known
Model form uncertainty	`NOT DONE`	The assumption of one temperature for the whole pack, an isothermal pack, was never tested against a model with several cells
Real hardware	`NOT DONE`	The whole course runs against a simulated pack
Sensor fault coverage	`PASS`	Bias, drift, lost readings (dropout), rounding (quantisation) and delay (latency) were all exercised
Reproducibility	`PASS`	Frozen configuration, fingerprint on every record of the log, fixed seeds

Five of the ten elements pass. The other five are the open items of the scorecard, and three of them concern uncertainty. Parameter uncertainty asks how well the data fix each parameter. Input uncertainty asks what happens when the inputs of the model, the current and the ambient temperature, are themselves measured with errors, while the twin treats them as exact. Model form uncertainty asks whether the equations are right: The model gives the whole pack one temperature, and this assumption was never tested.

Parameter uncertainty is only partial because of a symmetry that Day 3 found. Dividing the energy balance through by the heat capacity shows that the temperature depends on the four parameters only through three ratios:

$$\frac{dT}{dt} = \frac{R}{mc_p} I^2 - \left(\frac{hA_0}{mc_p} + \frac{hA_1}{mc_p} u \right) (T - T_{\text{amb}})$$

Multiplying R, $m\,c_p$, hA_0 and hA_1 by the same factor changes no ratio and therefore no temperature. Temperature data fix the three ratios and leave the common scale open. Intervals for the four single values are therefore not valid under this scaling symmetry. Day 3 computed such intervals with the bootstrap, by refitting the model to resampled data [9], and three of the four missed the true value. It then fixed the scale with a separate measurement of the heat capacity. Two parameter sets of this kind appear in the project of this day. The frozen configuration stores 0.0426 ohm, 8571 J/K, 2.05 W/K and 11.42 W/K, each about 0.93 times the calibrated value of the first table of this page, for example 0.0426 / 0.0456 = 0.934. The ratios of the two sets agree within 0.2 percent, and both give the same belief RMSE on the healthy pack, 0.174 °C.

The open items decide whether a twin may be deployed, and the row on real hardware weighs most. The whole course runs against a simulated pack, the usual way to develop a twin before hardware exists [10]. A twin validated only against its own simulator has shown that it is consistent with its assumptions, not that it corresponds to a battery.

> **Python code in the Colab notebook, Section 6.**
>
> Open Section 6 of the Colab notebook and run it. The list `CREDIBILITY` holds the ten rows of the table above. The cell prints the scorecard and counts the elements with the status `PASS` and the open items.

The assembled twin under a fault

The project ends with the complete twin, with its observer, its cumulative sum, its fail-safe policy and its computed maturity, run on a healthy pack and on a pack with a cooling fault and a raised internal resistance. One object collects the parts of the week and takes its settings from the frozen configuration. Each step of 10 s runs in a fixed order. The services of the twin compute the fan duty from the belief, an overheat flag and a trust flag. The overheat flag is set when the belief exceeds 46 °C, and the trust flag holds while the cumulative sum is below its decision limit. Then the fan command is chosen, the pack and the twin advance by one RK4 step, and once per minute a reading corrects the belief and updates the cumulative sum. A record of the step goes to the event log. The fan command follows the fail-safe rule

$$u = \begin{cases} \text{clip}\left((\hat{T} - T_{\text{on}})/6,\ 0,\ 1 \right) & \text{while } S < h \\ 1 & \text{otherwise} \end{cases}$$

A fail-safe policy is the action that is safe whatever the true state of the asset is. Here it is full cooling: It costs energy but can never make the pack hotter, so the twin chooses it as soon as it can no longer trust its own belief. At the end of a run, the report of the twin computes the maturity with the rules of the maturity section. Telemetry counts as present, and actuation counts as lost once an alarm was raised, because from then on the fan no longer follows the model of the twin. The report also gives the energy that the fan used, with 25 W at full duty:

> **In everyday life: a fail-safe action**
>
> Traffic lights whose controller fails switch to flashing amber, and a train brakes by itself when the driver releases the safety handle. In both cases the system cannot know what is happening, and it takes the action that is safe whatever the truth is. Full cooling plays this role for the pack.

$$E_{\text{fan}} = \frac{25 \times \Delta t \times \sum_k u_k}{3.6 \times 10^6}$$

The numerator is the energy in joules: the sum of the duties u_k of all steps, times 25 W and the step of 10 s. The denominator converts it, since 3.6×10^6 joules are one kWh. The faulted pack differs from the healthy one in three parameters. Its internal resistance is 1.35 times higher, $1.35 \times 0.045 = 0.06075$ ohm, its conductance without the fan is reduced to 45 percent, $0.45 \times 2.2 = 0.99$ W/K, and its fan adds only half, $0.5 \times 12 = 6.0$ W/K. The fault is present from the first second, and the twin is not told: It is the same object with the same configuration.

On the healthy pack the duties of the 540 steps add up to 187.8, a mean duty of 187.8 / 540 = 0.35. The formula gives $25 \times 10 \times 187.8 = 46950$ joules, and $46950 / (3.6 \times 10^6) = 0.0130$ kWh, the value in the report below. On the faulted pack the duties add up to 443.5, which gives $25 \times 10 \times 443.5 = 110\,875$ joules and $110\,875 / (3.6 \times 10^6) = 0.0308$ kWh.

Line of the report	Healthy pack	Faulted pack
Maturity	`DIGITAL TWIN`	`DIGITAL SHADOW`
Belief RMSE	0.174 °C	0.985 °C
Peak temperature of the pack	45.21 °C	49.87 °C
Time of the pack above 46 °C	0 s	4530 s
Time that the twin reported above 46 °C	0 s	4300 s
Energy of the fan	0.0130 kWh	0.0308 kWh
Trust lost at	never	1560 s
Records in the log	540	540

On the healthy pack, the twin keeps its trust and reports itself as a digital twin. On the faulted pack, the residual grows until the twin withdraws its trust after 1560 / 60 = 26 minutes and reports itself as a digital shadow. Because the cooling itself has failed, the pack still spends 4530 / 60 = 75.5 minutes above its limit, and the twin reports almost all of that time correctly: 4300 of the 4530 seconds. It reports no second in which the pack was below the limit. A twin cannot repair a broken asset, but it can say honestly what is happening.

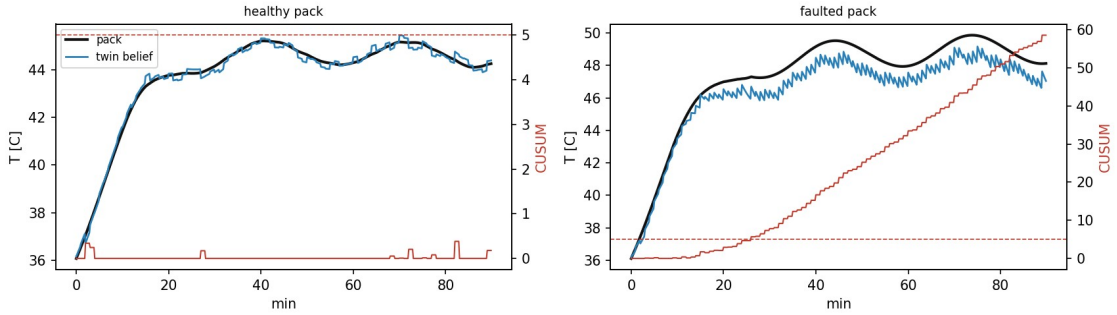


Figure. The assembled twin on the healthy pack, left, and on the faulted pack, right: the pack, the belief of the twin and, on the right axis, the cumulative sum with its decision limit.

In both panels the black line is the pack, the blue line the belief and the red line the cumulative sum, which belongs to the right axis. The dashed line is its decision limit of 5. On the healthy pack the belief follows the pack within a few tenths of a degree, and the sum never exceeds 0.4. On the faulted pack the temperature passes 46 °C after 14.5 minutes. The model of the twin expects more cooling than the pack

has, so between two readings the belief falls behind, and every reading pulls it up again, which gives the blue line its sawtooth shape. On average the belief lies 0.9 °C below the pack. The residuals of about 1.2 °C exceed the allowance, so the sum climbs, crosses its limit at minute 26 and reaches 58.5 at the end. From minute 26 on the fan runs at full speed, and the pack still peaks at 49.9 °C.

> **Worked example: why full cooling is not enough**
>
> The steady temperature T_∞ of the verification section tells where a pack settles. For the healthy pack at full fan and a current of 40 A, the heating is $40^2 \times 0.045 = 1600 \times 0.045 = 72$ W and the conductance $2.2 + 12.0 = 14.2$ W/K, so $T_\infty = 35 + 72 / 14.2 = 35 + 5.07 = 40.07$ °C, about 40.1 °C, far below the limit.
>
> For the faulted pack the heating is $1600 \times 0.06075 = 97.2$ W and the conductance $0.99 + 6.0 = 6.99$ W/K, so $T_\infty = 35 + 97.2 / 6.99 = 35 + 13.91 = 48.91$ °C, about 48.9 °C. Even with the fan at full speed from the first second, this pack settles above its limit of 46 °C. No control policy can prevent the overheating at this load. Only a lower current or a repair can.

Part E of the interactive lab runs such a twin in the browser for two hours, with a fault that the reader injects at a chosen minute. The fault is a fan that loses 70 percent of its effect, a sensor that reads 4 °C too low, or a blockage that halves the cooling without the fan. The panel differs from the twin of this page in one design decision, which answers a weakness of every twin that corrects its belief with the readings: Such a belief also follows the errors of the sensor. The dashboard shows this weakness under a sensor bias of −2 °C: The belief settles 1.9 °C below the true temperature. The residuals shrink to a few tenths of a degree, and the cumulative sum stays silent. The pack then spends 2010 seconds above its limit, and the twin reports none of them.

In the panel the belief copies a reading once per minute, so it follows a biased sensor completely. The panel therefore keeps a second temperature next to the belief: a physics prediction T_{phys} that is computed from the model alone and never corrected by a reading. The residual of the trust check is the reading minus this prediction, and the panel averages the last 30 of them, the readings of five minutes:

$$\bar{r} = \frac{1}{30} \sum_{j=1}^{30} \left(y_j - T_{\text{phys},\, j} \right), \qquad \text{trust is lost when } |\bar{r}| > 1$$

The bar marks a mean, the index j runs over the last 30 readings, and the limit of 1 is in degrees Celsius. A biased sensor shifts every reading but not the physics prediction, so the mean residual moves by the full bias and the fault is found. Detection needs a reference that does not depend on the sensor under suspicion. The price is that the physics prediction drifts when the model is slightly wrong, which is why the limit is 1 °C and the mean covers five minutes. A lower limit or a shorter window reacts faster but raises more false alarms.

> **Interactive lab, Part E.**
>
> The operations panel: the assembled twin over two hours in the browser, with a fault that you inject at a chosen minute, a weaker fan, a biased sensor or a blocked cooling path, and the physics check that finds a biased sensor. Part E opens at https://utkukose.github.io/digital-twin-veltech-NB-lecture/days/day-05/lab.html#pe.

Other faults are staged in the same way as the fault above. A fault of the asset is another parameter set for the pack, for example a fan conductance of zero for a fan that is stuck. A fault of the sensor is another description of the sensor, for example with a bias. The daily task asks for two such runs.

Python code in the Colab notebook, Section 7.

Open Section 7 of the Colab notebook and run its cells. The class `DigitalTwin` is the assembled twin: Its method `run` drives one trip and `report` returns the lines of the table above. The first cell runs it on the healthy pack `TRUE` and prints the report. The second cell runs it on the faulted pack `DEGRADED`, prints the second report and the time at which trust was lost, and draws the figure of this section. To stage another fault, pass another `PackParams` to `run`, for example with `hA1=0.0` for a fan that is stuck.

Check your understanding (5). The cooling of the pack fails and the pack overheats although the twin works correctly. What is the value of the twin here?

A. None

B. It detects the fault, withdraws its trust and reports the overheating honestly

C. It repairs the fan

D. It hides the fault

The twin of the pack, stage 5: a twin that knows how far it can be trusted

The methods of this day complete the project. The parts of the week now run as one object with a frozen configuration, an observer, a trust check and a fail-safe fan command, and every step leaves a record in the event log. The first figure shows the last stage of the project, the second the assembled twin.

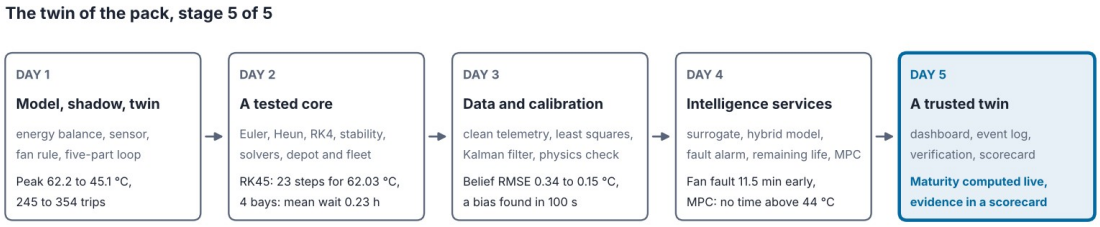


Figure. The stage of the project reached today: the assembled twin with its evidence of trust.

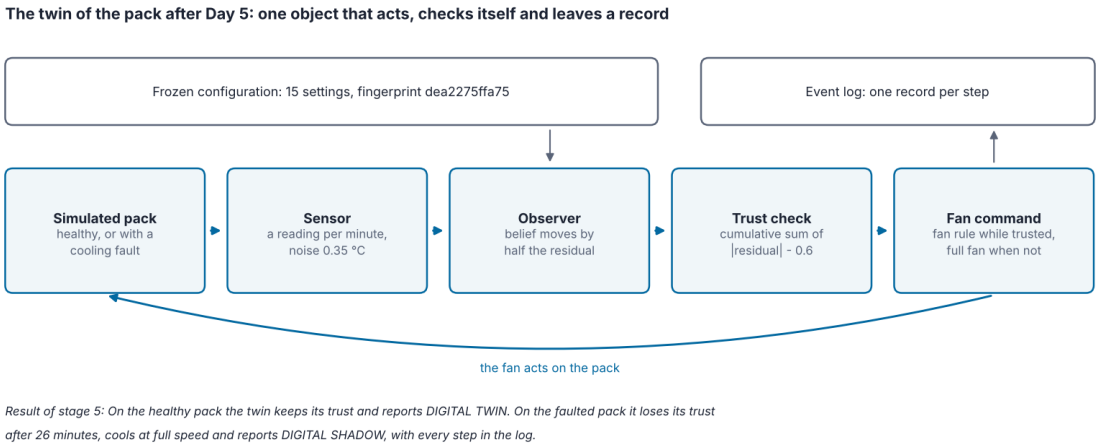


Figure. The assembled twin of the pack. The configuration sets every part of the loop, and the log keeps every step with its fingerprint.

Question	Answer of stage 5	Section
What does the operator see?	The temperature against its limit, the fan duty and the cumulative sum that carries the trust	A dashboard for the operator
Is the label true right now?	The maturity is computed from telemetry, actuation and the error of the belief	Maturity computed from the state of the twin
Can a run be repeated and checked?	A frozen configuration with the fingerprint dea2275ffa75 in every record of the log	Configuration and the event log
Are the equations solved right, and are they right?	Observed order 4.02, discretisation error 0.0002 °C, and validation down to the noise floor of 0.35 °C	Verification and validation
How far can it be trusted?	Five of ten elements of the scorecard pass, and a test on real hardware is still missing	The credibility scorecard
What happens under a fault?	Trust is lost after 26 minutes, the fan runs at full speed, and 4300 of 4530 hot seconds are reported	The assembled twin under a fault

The five stages together are the twin of the course: a model of the pack, a tested simulation core, clean data with calibrated parameters and a filtered belief, a layer of services and an assembled twin that knows how far it can be trusted. The final project below asks for the same five stages for an asset of the student's own choice.

> **Python code in the Colab notebook, Section 8.**
>
> Open Section 8 of the Colab notebook and run its cells. The function `live_dashboard` shows the assembled twin of section 7 as a live page, so the answers of the table above can be followed while a trip runs. The pack, its sensor and its fan are animated, the maturity and the trust checks are computed at every step, and every record of the event log carries the fingerprint of the configuration. Faults of the pack, of the sensor and of the two data flows can be staged at any minute. Besides the cumulative sum, the trust rests on the physics check of Day 3 and on a check of the telemetry. At the end of a trip without staged faults, the page compares its report with the report of `DigitalTwin`.

Your asset and the final project

The application of the day builds a small operator view and a scorecard of four checks for four assets, and a worn fan can be staged without telling the twin. A live page then runs the chosen asset step by step and updates the four checks while the run goes on. The four assets are a battery pack, a server rack, a motor winding and the air of a greenhouse. They share the energy balance of the pack, with a heat input $P(t)$ in watts in place of $I^2 R$ and a heat capacity C in place of $m\,c_p$. The section integrates it with the explicit Euler method:

$$T_{k+1} = T_k + \Delta t \, \frac{P(t_k) - (hA_0 + hA_1 u_k)\,(T_k - T_{\mathrm{amb}})}{C}$$

Here k counts the steps. The duty u_k follows the rule of the dashboard with a threshold 6 °C below the limit of the asset, so the fan reaches full speed at the limit. The readings carry a noise with a standard deviation of one percent of the distance between the ambient temperature and the limit, $\sigma = 0.01\,(T_{\mathrm{limit}} - T_{\mathrm{amb}})$.

Asset	Heat capacity C in J/K	hA_0 and hA_1 in W/K	Ambient and limit in °C	Length of the run and step	Noise σ in °C
Battery pack	9000	2.2 and 12	35 and 46	2 hours, 5 s	0.11
Server rack	120 000	40 and 400	24 and 35	2 hours, 10 s	0.11
Motor winding	4000	1.5 and 8	30 and 90	1.5 hours, 5 s	0.60
Greenhouse air	3 000 000	500 and 4000	18 and 32	24 hours, 60 s	0.14

The scorecard has four checks. The verification check estimates the order of the integrator without an exact solution, from three runs with the steps Δt, $\Delta t / 2$ and $\Delta t / 4$:

$$p_{\text{obs}} = \log_2 \left(\frac{\max_k |a_k - c_k|}{\max_k |b_k - c_k|} - 1 \right)$$

a_k, b_k and c_k are the temperatures of the three runs at the same times, from the coarsest to the finest, and $\max_k$ takes the largest difference over the run. For a method of order p the fraction equals $2^p + 1$, so the formula returns p: A fraction of 3 gives $3 - 1 = 2$ and $\log_2 2 = 1$, the order of the Euler method. The check passes when the result lies within 0.3 of 1. The validation check passes when the RMSE between the twin and the readings stays below $2\,\sigma$. The two operation checks pass when the peak temperature stays below the limit and no time is spent above it.

For the battery pack without wear, all four checks pass. The observed order is 1.00, the RMSE of 0.110 °C equals the noise, and the peak of 43.8 °C stays below the limit of 46 °C. With the worn fan, the real cooling loses half of the effect of its fan while the twin keeps its old model. The twin of this section is the model alone, which no reading corrects, so the wear appears as a difference between the twin and the readings. For the battery pack, the validation check then fails, with an RMSE of 1.53 °C against a noise of 0.11 °C. The pack itself stays below its limit, with a peak of 45.6 °C, so the scorecard reports the wear before any limit is reached. For the motor, the same wear passes all four checks. Its fan runs at about a tenth of full speed on average, so halving its effect changes the temperature little, and the RMSE of 0.89 °C stays below twice the noise, $2 \times 0.60 = 1.20$ °C. A check can only be as sharp as its sensor.

> **Python code in the Colab notebook, Section 9.**
>
> Open Section 9 of the Colab notebook. Its first line is a switch, `ASSET = "battery"`. Replace the word by `server_rack`, `motor` or `greenhouse`, set `WORN_FAN = True` to wear out the fan, and run the section. The function `assess_asset` runs the asset and its twin and fills the four checks. The cell prints the scorecard with the value of every check and draws the readings, the twin and the limit above the cooling duty. Exercise B asks for the number of checks that pass. At the end of the section, `live_asset` runs the same asset and its twin as a live page, where the worn fan, the noise of the readings and the heat input can be changed during the run.

The final project then asks every student for a twin of a freely chosen asset, with every element of the week. Its brief asks for an asset with at least two coupled states and one controllable input. The twin needs both data flows, a tested simulation core, calibrated parameters, two elements of the intelligence layer, a dashboard and a filled scorecard. Each requirement was built on one day of the course. The data flows are those of Day 1 and of the dashboard, and the tested core is the verification of Day 2 and of this day. The calibration includes the identifiability check of Day 3, the question whether the data can fix each parameter. The intelligence layer is the set of services of Day 4, such as a surrogate, a fault detector or a planning controller. The scorecard must name at least two open items with their evidence.

> **Python code in the Colab notebook, Section 10.**
>
> Open Section 10 of the Colab notebook and read the brief of the final project and its rubric. This section has no code to run.

Answers to the knowledge checks

Quick check. So that no setting can change silently while the twin runs: Every change requires a new object with a new fingerprint, which then appears in the log.

1. B. The absolute value of the residual counts: 4.2 plus 1.5 minus 0.6 gives 5.1, which exceeds the decision limit of 5.0.

2. B. Maturity is a property of the running system, not of the design document.

3. B. Below the noise floor, the data cannot distinguish a small model error from none.

4. B. The squares are subtracted, not the values: 0.0169 minus 0.0144 is 0.0025, and its square root is 0.05 °C.

5. B. Detection and honest reporting are what a twin can offer when the asset itself fails.

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